

Bandwidth Enhancement of Bow-tie Microstrip Patch Antenna Using Defected Ground Structure for 5G

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Abstract—A microstrip antenna suffers from a narrow bandwidth because of the thickness of the substrate. This paper presents a wideband microstrip antenna by using Defected Ground Structure (DGS) on a bow-tie microstrip patch antenna. The cross-dumbbell shaped slot is used as DGS on the ground layer. The cross-dumbbell DGS slot is put on the backside of the transmission line. The antenna is designed using Computer Simulation Technology (CST) microwave studio and on the commercially printed circuit board substrate, i.e., the FR-4 substrate. The antenna is set for 5G applications, which use 3.67 GHz as the frequency center. By using DGS, the simulated reflection coefficient (S_{11}) for $VSWR \leq 2$ shows the fractional bandwidth increase from 2.6% to 10.45%. The measured reflection coefficient matches the simulated results where the measurement gives 13.37% for fractional bandwidth.

Index Terms—Bandwidth enhancement, defected ground structure, Bowtie patch, 5G antenna, microstrip antenna

I. INTRODUCTION

In recent years, microstrip antennas have been widely used for many applications, including for 5G applications [1]–[5]. The microstrip antenna can be used on the Base Transceiver Station (BTS). The microstrip antenna has a light weight, low profile configuration, conformal structure, reliability, ease of fabrication process and low cost of fabrication [6]. However, the microstrip antenna has drawbacks such as a narrow impedance bandwidth and low gain. The antenna gain is associated with the dielectric specification. Antenna gain increases as the height increases and degrades as the permittivity increases. [7].

Many reports offer their studies to enhance impedance bandwidth. As reported in [8], resulting electromagnetic coupling on the stacked substrates can improve impedance bandwidth. The stacked substrate decreases the quality factor of the patch antenna (Q) more than it improves impedance bandwidth [9]. However, alignment of the stacked substrate must be a concern to give better measurement results.

Defected Ground Structure (DGS) is an emerging technique for improving narrow bandwidth, high selectivity, and low gain on microwave circuits [10]. DGS

can be used in antennas [11]–[13], filters [14]–[16] and other microwave components. [11], [12] use circular and rectangular shapes of DGS to disturb the electric field on the ground layer. However, fractional bandwidth improvement is still limited to 12% [11].

Bow-tie patch modifications can also be used to increase bandwidth [17]–[20]. Impedance bandwidth was increased by 10.3% by using a concentric ring slot bow-tie patch in conjunction with DGS [18]. A bow-tie patch with a slant slot is implemented at 27.77 GHz and it achieved a 6.3% impedance bandwidth measurement [20]. [21] used a bow-tie slot on the rectangular Substrate Integrated Waveguide (SIW) antenna to generate hybrid resonant frequencies. Hybrid resonant frequencies occur because of a combination between stronger or weaker TE_{110} and TE_{120} modes. The same method has been successfully implemented on the circular SIW antenna as shown in [22]. However, fractional bandwidth achievement is still below 10%.

This paper presents impedance bandwidth enhancement by using a bow-tie microstrip patch antenna that is combined with a cross-dumbbell DGS slot. The cross-dumbbell DGS slot has been successfully implemented into a low-pass filter with an ultra-wide stopband [14]. The novel implementation of cross-dumbbell DGS into bow-tie microstrip patch antenna improves up to 13.37% fractional bandwidth. The antenna design uses an FR-4 substrate with a 1.6 mm thickness. The antenna design works on 3.5 GHz, which can be implemented into 5G applications.

II. MICROSTRIP ANTENNA DESIGN

A. Dimension of Patch Bowtie

The shape of a butterfly-structured antenna, also known as a bow-tie antenna, is designed by the basic shape of the triangular (triangular) shape [23] in the symmetric magnetic wall. Naturally, the bow-tie patch antenna also offers a more compact nature than the rectangular patch antenna [6]. While physically, the triangular shape has the same radiation properties as the rectangular patch, the radiation losses are smaller than the rectangular patch shapes. Therefore, the bow-tie patch antenna is usually used to enhance impedance bandwidth [24].

FR-4, an epoxy substrate is used for antenna design, with a 1.6 mm substrate thickness (h) and a 4.3 dielectric constant (ϵ_r). The general specification of the bow-tie antenna dimension is shown in Fig. 1. The general specifications of bow-tie antennas, such as the width (W) and the length (L), are similar to those of rectangular patch dimensions with some shape modifications. The width of a rectangular patch antenna can be calculated by using the following equation [6], [25]:

$$W = \frac{c}{2f \left(\sqrt{\frac{\epsilon_r + 1}{2}} \right)} \quad (1)$$

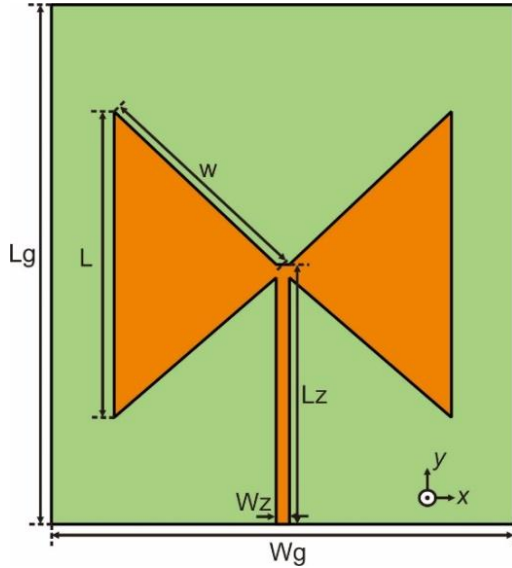


Fig 1. The geometrics of the bow-tie patch antenna.

While the rectangular patch antenna's length is determined by equation:

$$L = \frac{c}{2f \sqrt{\epsilon_{reff}}} - 2 \Delta L \quad (2)$$

The *effective dielectric constant* (ϵ_{reff}) can be calculated by using the following equation:

$$\epsilon_{reff} = \frac{(\epsilon_r + 1)}{2} + \frac{(\epsilon_r - 1)}{2} \left[1 + 12 \frac{h}{W} \right]^{-\frac{1}{2}} \quad (3)$$

and the *Length Extension* (ΔL) can be determined by this equation:

$$\Delta L = 0.412h \frac{(\epsilon_{reff} + 0.300) \left(\frac{W}{h} + 0.262 \right)}{(\epsilon_{reff} - 0.258) \left(\frac{W}{h} + 0.813 \right)} \quad (4)$$

The effective length (L_{eff}) is calculated by using:

$$L_{eff} = \frac{c}{2f \sqrt{\epsilon_{reff}}} \quad (5)$$

The feeding transmission line technique in the microstrip antenna is very important to give good impedance matching. The feeding transmission line uses a 50 ohm for the transmission line width (Wz) and a half of the width that the PCB is used for the transmission line length (Lz). The transmission line width can be calculated by this equation [26]:

$$A = \frac{Z_0}{60} \sqrt{\frac{\epsilon_r + 1}{2}} + \frac{\epsilon_r - 1}{\epsilon_r + 1} \left(0.23 + \frac{0.11}{\epsilon_r} \right) \quad (6)$$

where A is the logarithmic constant

$$\frac{Wz}{h} = \frac{8e^A}{e^{2A} - 2} \quad (7)$$

TABLE I: ANTENNA PARAMETERS

Parameter	Dimension (mm)	Description
W	26.3269	width of a triangle
L	20.2168	length of a triangle
h	1.6	substrate thickness
t	0.1	copper thickness of the substrate
Wg	90	the substrate width
Lg	80	the substrate length
Wz	3.1118	transmission line width
Lz	45	transmission line length

The optimization of the bow-tie microstrip patch antenna dimension is tabulated in Table I .

B. Cross-Dumbbell Defected Ground Structure Slot

Surface wave propagation is a serious problem for microstrip antennas. Surface waves reduce gain efficiency, narrow bandwidth, increase end-fire radiation, increase cross-polarization rate and limit frequency application. The electromagnetic band gap (EBG) can be offered to overcome this phenomenon. The development of EBG is DGS, which has been widely used in other microwave components. DGS can suppress surface waves and it is often implemented on the microstrip antenna. [10]

There are many kinds of DGS shapes, such as dumbbell shapes with various heads, hilbert curve ring, meander and many more. This antenna design uses a pair of rectangular dumbbell shapes that are placed perpendicular to each other. It is named by a cross-dumbbell DGS slot as shown in Fig. 2. [14]

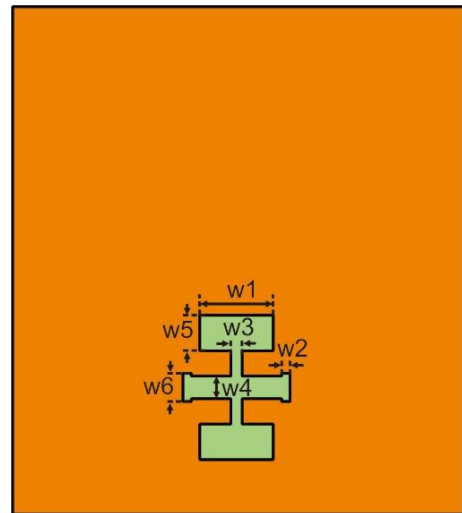


Fig. 2. The cross-dumbbell DGS.

By using the CST studio suite, the antenna design is simulated and optimized for the frequency application. The optimization of the cross-dumbbell DGS slot dimension is listed in Table II. Fig. 3 shows the reflection coefficient simulation for antenna design without and with

a cross-dumbbell DGS slot. It is shown that the cross-dumbbell DGS slot enhances impedance bandwidth simulation. The microstrip patch antenna without a cross-dumbbell DGS slot has a 2.66% (3.46–3.55 GHz) fractional bandwidth, while the antenna design with a cross-dumbbell DGS slot has a 10.44% (3.45–3.83 GHz) fractional bandwidth. The cross-dumbbell DGS slot has improved fractional bandwidth to 3.93 times that of the bow-tie microstrip patch antenna. The antenna design with cross-dumbbell DGS simulation has three resonance frequencies that occur at 3.5, 3.66, and 3.77 GHz.

TABLE II: CROSS DUMBBELL DGS PARAMETER

Parameter	Dimension (mm)
W1	13
W2	1.5
W3	2
W4	4
W5	6.3
W6	5

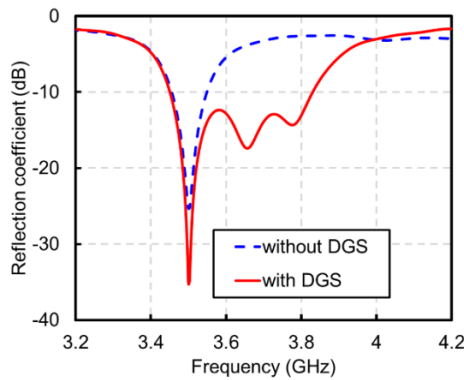


Fig. 3. Reflection coefficient simulation without DGS and with DGS.

C. Parametric Studies Cross-Dumbbell Defected Ground Structure

The cross-dumbbell DGS is placed on the ground plane, which is placed on the back side of the transmission line. The cross-dumbbell DGS has six parameters that influence impedance bandwidth enhancement. The six parameters of cross-dumbbell DGS are i.e. W1, W2, W3, W4, W5, and W6.

The cross-dumbbell DGS on the ground plane successfully enhances impedance bandwidth by combining three resonance frequencies in the reflection coefficient range below -10 dB. A parametric study for W1 parameters is shown in Fig. 4. The lengthened W1 parameter caused three resonance frequencies to shift into the lower frequency. We observed that for W1 = 11 mm, the dual-band frequency occurs on this antenna design with 4.51% (3.47–3.63 GHz) and 2.85% (3.8–3.91 GHz) fractional bandwidth. The dual-band frequency also occurs for W1 = 15 mm with a resulting 5.39% (3.43–3.62 GHz) and 2.14% (3.70–3.78 GHz) fractional bandwidth. The best W1 parameter for this antenna design is obtained when W1 = 13 mm with 10.44% fractional bandwidth.

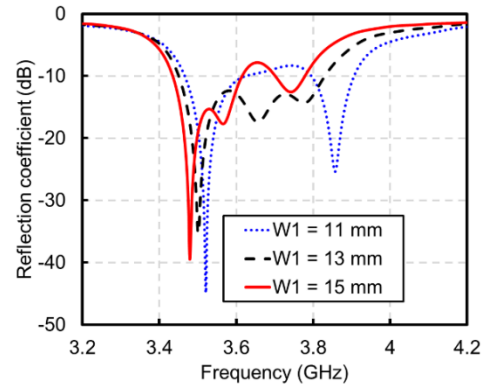


Fig. 4. Variation of reflection coefficient with different W1 parameters.

Impedance bandwidth can also be seen by the standing wave ratio (VSWR) parameter with a ≤ 2 level, as shown in Fig. 5. Fig. 5 shows the variation of the standing wave ratio (VSWR) with different W1 parameters. According to standing wave ratio ≤ 2 level, it is shown that the W1 = 13 mm has a wide impedance bandwidth rather than other W1 parameters.

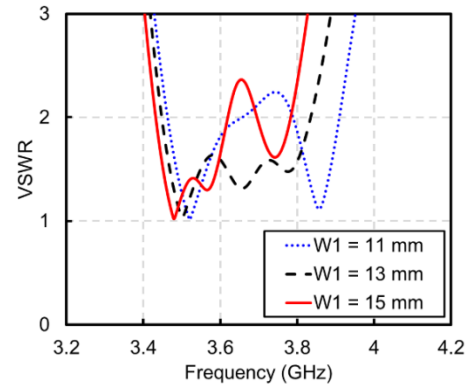


Fig. 5. Variation of standing wave ratio with different W1 parameters.

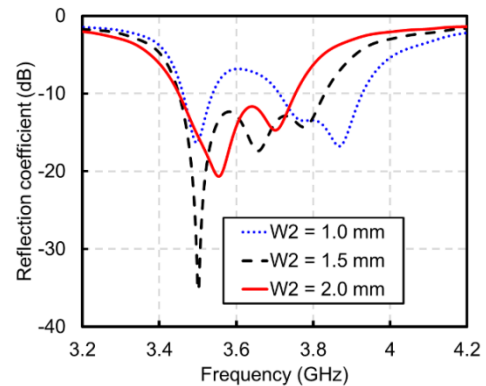


Fig. 6. Variation of reflection coefficient with different W2 parameters.

Fig. 6 shows the influence of the W2 parameter. We observed that the thickened W2 parameter caused the antenna design to have a narrow impedance bandwidth with dual resonance frequencies. It occurs because three resonance frequencies disappear and become dual resonance frequencies. When W2 = 1 mm, the dual-band frequency occurs with 2.29% (3.46–3.54 GHz) and 5.76%

(3.71–3.93 GHz). The dual-band frequency with three resonance frequencies merges together into a wideband frequency until finally the three resonance frequencies are reduced to a narrow bandwidth. When the W2 parameter is set to 2.00 mm, the fractional bandwidth is 8.33% (3.45–3.75 GHz).

The standing wave ratio (VSWR) with different W2 parameters is shown in Fig. 7. For W2 = 1.5 mm, the standing wave ratio has ≤ 2 levels lower than the other W2 parameters. It occurs because of the dual-band frequencies merging together to become three resonance frequencies.

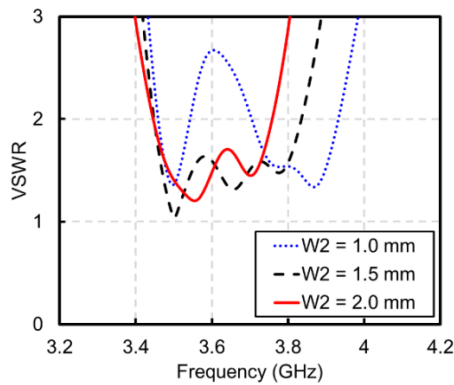


Fig. 7. Variation of standing wave ratio with different W2 parameters.

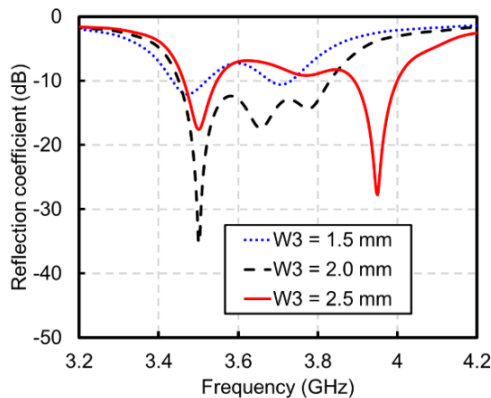


Fig. 8. Variation of reflection coefficient with different W3 parameters.

In Fig. 8, the reflection coefficient for different W3 parameters is shown. We recommend that the W3 width be no more than the transmission line width. The thickened W3 parameter caused three resonance frequencies to separate and the antenna design to become a dual-band antenna. The fractional bandwidth enhancement of this antenna design is achieved when W3 = 2.0 mm. It has three resonance frequencies with a 10.44% (3.45–3.83 GHz) fractional bandwidth, but when W3 = 2.5 mm, the antenna design becomes a dual-band antenna with 2.57% (3.46–3.55 GHz) and 2.79% (3.89–4.00 GHz) fractional bandwidths.

Fig. 9 shows the standing wave ratio (VSWR) with different W3 parameters. For W3 = 1.5 mm and 2.5 mm, the standing wave ratio is higher than the 2 magnitude level. It occurs because the W3 width slots are too thin or too thick.

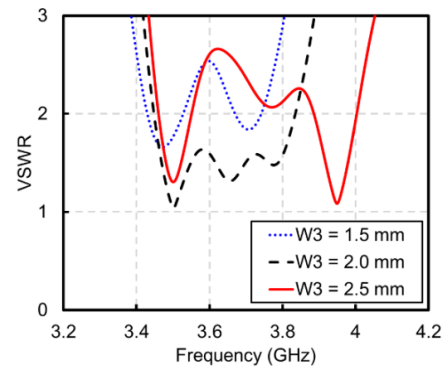


Fig. 9. Variation of standing wave ratio with different W3 parameters.

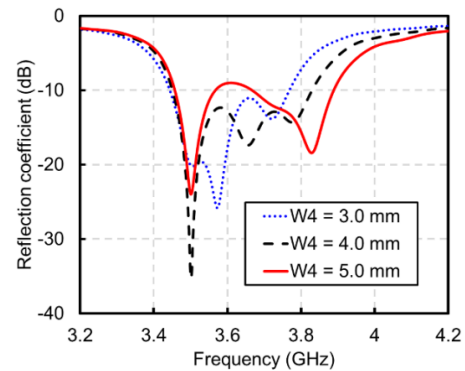


Fig. 10. Variation of reflection coefficient with different W4 parameters.

The reflection coefficient with different W4 parameters is shown in Fig. 10. The first resonance frequency is more stable than the second and third resonance frequencies. The thickened W4 parameter caused the second and third resonance frequency shifts into the higher frequency. For example, the second resonance frequency shifts from 3.57 GHz into 3.66 GHz and 3.73 GHz when W4 parameters are 3.0, 4.0, and 5.0 mm. The third resonance frequency also shifts from 3.73 GHz to 3.78 GHz and 3.83 GHz for the thickened W4 parameter. The shifting of the second and third resonance frequencies caused the impedance bandwidth to widen for certain W4 parameters. The optimization of W4 parameters for this antenna design occurs at W4 = 4.0 mm.

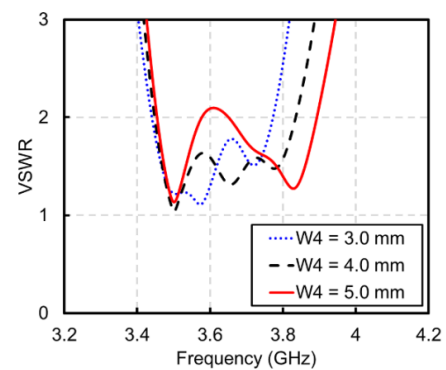


Fig. 11. Variation of standing wave ratio with different W4 parameters.

The standing wave ratio (VSWR) with different W4 parameters is shown in Fig. 11. The first resonance

The standing wave ratio (VSWR) with different W_4 parameters is shown in Fig. 11. The first resonance frequency is more stable than the second and third resonance frequencies. The wide impedance bandwidth optimization occurs for $W_3 = 4.0$ mm when the standing wave ratio is lower than the 2 magnitude level.

Fig. 12 shows the reflection coefficient with different W_5 parameters. We observed that the changes of the thicken W_5 parameters are reciprocal with W_4 parameters, where the second and third resonance frequencies shift into the lower frequency for the thicken W_5 parameter. For certain values of the W_5 parameter, the wideband impedance bandwidth with three resonance frequencies is achieved. It occurs because of the change from the dual-band frequency to merging three resonance frequencies on the reflection coefficient (S_{11}) below -10 dB.

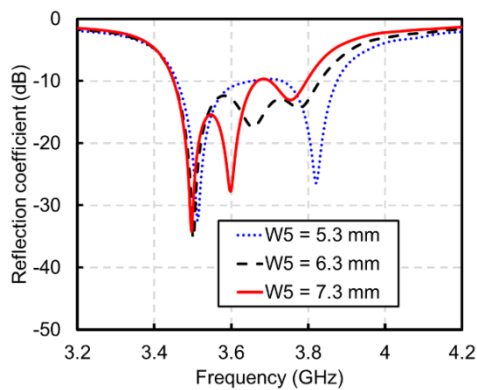


Fig. 12. Variation of reflection coefficient with different W_5 parameters.

Fig. 13 shows the VSWR with different W_5 parameters. The thinner W_5 parameters caused the second resonance frequency to disappear and become a dual-band frequency. The wide impedance bandwidth optimization occurs for $W_5 = 6.3$ mm.

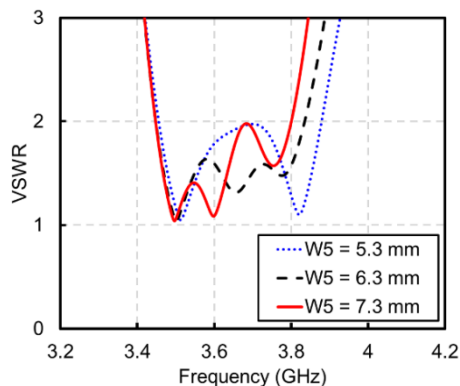


Fig. 13. Variation of standing wave ratio with different W_5 parameters.

The reflection coefficient with different W_6 parameters is shown in Fig. 14. The shifting of the second and third resonance frequencies is influenced by the thickened W_6 parameter. They shift to a lower frequency. For certain W_6 parameters, there are three resonance frequencies that combine together to cause impedance bandwidth improvement. For example, the impedance bandwidth

with three resonance frequencies is achieved for $W_6 = 5.0$ mm and $W_6 = 5.5$ mm with 10.44% (3.45–3.83 GHz) and 9.66% (3.45–3.8 GHz). The antenna design for $W_6 = 4.5$ mm has dual-band frequency with 3.42% (3.45–3.57 GHz) and 6.68% (3.62–3.87 GHz) fractional bandwidth. The dual-band frequency antenna occurs because of the second and third resonance frequencies shifting into the higher frequency when the W_6 width decreases.

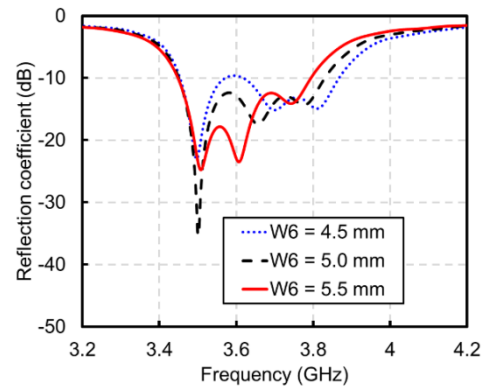


Fig. 14. Variation of reflection coefficient with different W_6 parameters.

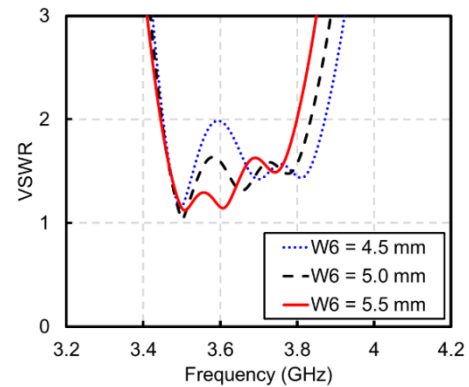


Fig. 15. Variation of standing wave ratio with different W_6 parameters.

The standing wave ratio (VSWR) with different W_6 parameters is shown in Fig. 15. The dual-band frequency merges into a wide impedance bandwidth when $W_6 = 5.0$ mm. Impedance bandwidth is narrow when the W_6 width slot is thicker than others.

III. RESULT AND DISCUSSION

Fig. 16 shows the photographs of the fabricated antenna design. The fabricated antenna design used a photo etching process. The fabricated antenna design makes use of an FR-4 substrate, which is less expensive than other substrates, lightweight, and widely available.

The reflection coefficient measurement is done to validate the ratio of the reflected wave that compares with the transmitted wave. The reflected waves occur due to a mismatch in impedance bandwidth mismatch between the load (antenna) and the transmission line. The reflection coefficient measurement used Vector Network Analyzer equipment at the Research Centre for Electronics and Telecommunications (BRIN)-Bandung.

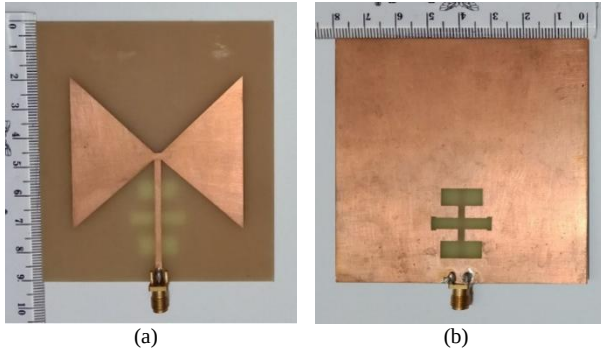


Fig. 16. The designed and built antennas for (a) the bow-tie patch and (b) the cross-dumbbell DGS slot.

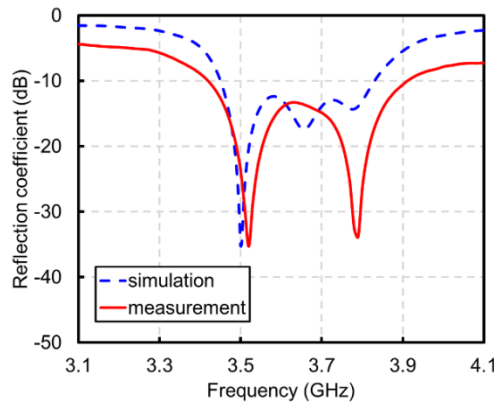


Fig. 17. DGS was used to compare the reflection coefficients of measurement and simulation fabricated antennas.

Fig. 17 shows the reflection coefficient comparison between simulation and measurement. The measured fabricated antenna used a cross-dumbbell DGS slot while the cross-dumbbell DGS slot dimension is tabulated in Table II. The measured reflection coefficient occurs at 3.42–3.91 GHz, which means the antenna design has 13.37% fractional bandwidth. The differentiation between simulated and measured reflection coefficients occurs because of the changing dimensions of the antenna that happen in the fabrication process. The measured three resonance frequencies occur at 3.52, 3.7, and 3.79 GHz.

Fig. 18 shows the radiation pattern simulation for three resonance frequencies: 3.5, 3.66, and 3.77 GHz. The antenna design has a unidirectional radiation pattern for the YOZ plane. The gain antenna simulated was 8.38 dBi at 3.5 GHz as shown in Fig. 19. The antenna design has an omnidirectional radiation pattern in the XOZ plane. Table III shows a comparison between this antenna design result and other antenna designs.

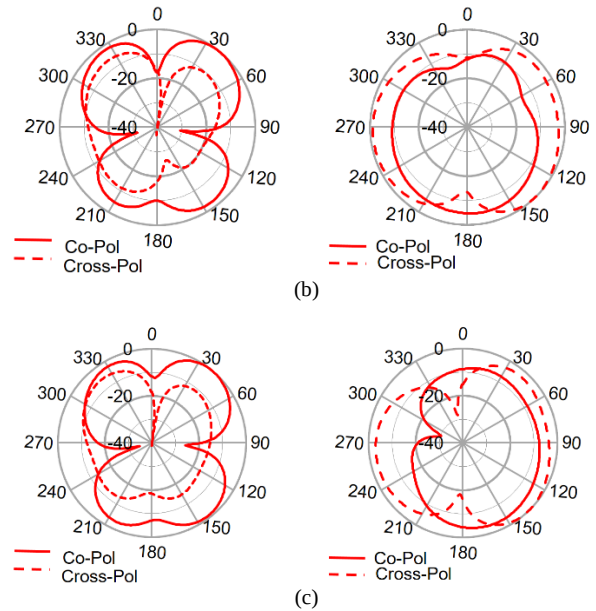
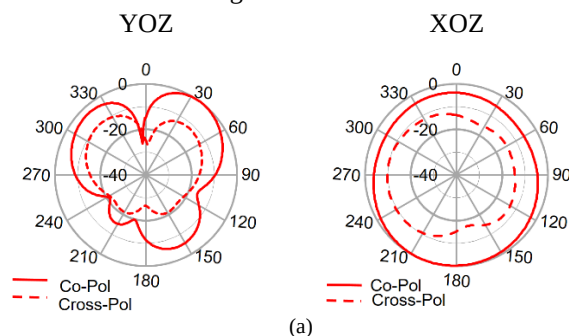


Fig. 18. (a) 3.5 GHz, (b) 3.66 GHz, and (c) 3.77 GHz radiation patterns

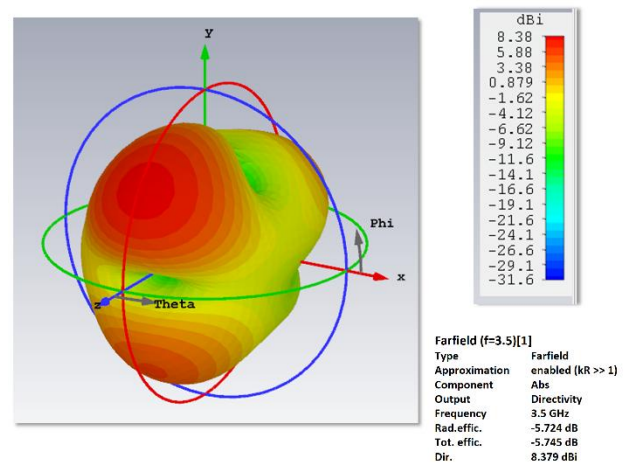


Fig. 19. Simulated gain antenna at 3.5 GHz.

TABLE III: COMPARISON OF SIMULATION AND MEASUREMENT RESULT

Reference	Substrate	BW (GHz)	Fc (GHz)	FBW (%)
[18]	FR-4	1.5	14.25	10.52
[20]	RO-5880	1.75	27.77	6.3
[22]	RO-5880	0.22	8.45	2.6
This Work	FR-4	0.49	3.67	13.37

IV. CONCLUSION

A bow-tie microstrip patch antenna with a defected ground structure (DGS) slot was proposed in this paper to enhance impedance bandwidth by more than 10%. The DGS type was a cross-dumbbell slot shape, and it was put on the ground plane. The antenna design was fabricated using an FR-4 substrate and the measured impedance bandwidth for the fabricated antenna was achieved at 13.37%. The frequency of this antenna was available for 5G applications at 3.42–3.91 GHz.

CONFLICT OF INTEREST

The authors declare no conflict of interest

AUTHOR CONTRIBUTIONS

All of the author's contributions are in this paper. Dian Widi Astuti conducted this research; Rizki Fadilah simulated the antenna design; Muslim fabricated the antenna design; Dian Rusdiyanto analyzed the data; Syah Alam wrote the paper; and Yuyu Wahyu measured the fabricated antenna.

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